LUQER

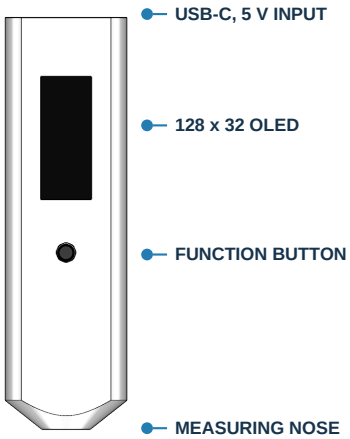
DIGITAL MAGNETIC THICKNESS GAUGE

POCKET MANUAL

Firmware AL 1.7.20

English - two-sided A4 fold edition

PROBE AT A GLANCE 02



TECHNICAL SPECIFICATIONS 03

MEASURING RANGE	0.00-7.50 mm
DISPLAY RESOLUTION	0.01 mm
ZERO DISPLAY CLAMP	+/- 0.03 mm may show 0.00
TARGET REPEATABILITY	+/- 0.04 mm, stable conditions
BODY DIMENSIONS	85.3 x 25.2 x 17.0 mm
POWER	USB Type-C, 5 V
DISPLAY / CONTROL	128 x 32 OLED / one button
SUPPLIED MAGNET	10 mm diameter sphere

The firmware displays -/- beyond approximately 7.60 mm. Firmware reference: AL 1.7.20.

PACKAGE + POWER 04

IN THE PACKAGE

- LUQER measuring probe
- 10 mm magnetic sphere
- Fitted storage case

NOT SUPPLIED

A USB Type-C cable and 5 V USB power source are required for operation but are not included in the package.

LOW-CURRENT NOTE

Some power banks automatically switch off when the connected device draws very little current. This does not necessarily indicate a fault in the instrument.

STRONG MAGNET SAFETY 05

10 mm MAGNET SPHERE

The supplied sphere is a strong magnet. It can jump, pinch skin, chip, or become a projectile. Wear eye protection when the work creates an impact risk.

- Keep away from children and never place in the mouth.
- Keep away from pacemakers, implanted medical devices, cards, watches, magnetic media, and sensitive electronics. Follow device-maker separation guidance.
- Do not use a cracked, corroded, loose, heated, drilled, or ground magnet.
- Store the sphere in the case's dedicated magnet storage slot whenever it is not in use. Never leave it loose on a bench or elsewhere in the case.

USE ON FINISHED WORK 06

SURFACE PRECAUTIONS

The sphere can snap, slide, or trap hard particles against the work. Protect delicate finishes and test the method on a non-critical area first.

- Use clean protective film only if the measurement method allows it; added thickness changes the result.
- Do not drag the sphere or probe nose across the surface.
- Confirm that magnetic force will not damage coatings, thin parts,
- electronics, or internal mechanisms. The operator is responsible for suitability of the method and
- independent verification of critical results.

PERFORMANCE STATUS 07

TARGET - NOT YET VALIDATED

Target repeatability is +/- 0.03 mm under stable conditions. This is a product goal, not a validated claim.

Accuracy depends on calibration, magnet position, geometry, protective layers, and temperature. Verify critical readings against known standards.

Display resolution is 0.01 mm; this does not establish accuracy or measurement uncertainty.

Before acceptance work:

- Verify with traceable standards across the intended range.
- Establish repeatability, reproducibility, environmental effects, and uncertainty.

RECORD + DISCLAIMER 08

RECORD WITH THE INSTRUMENT

- Serial number
- Firmware version AL 1.7.20
- Calibration and verification dates
- Reference standards used
- Observed damage or repairs

IMPORTANT

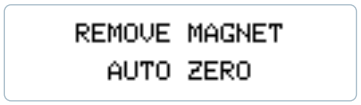
LUQER is a comparative measuring instrument. Do not use it as the sole basis for safety-critical acceptance. Suitability, traceability, and measurement uncertainty must be established by the user.

Specifications in this manual reflect supplied information and firmware AL 1.7.20

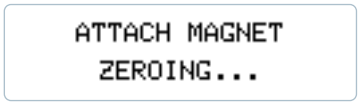
START-UP 09

Connect regulated 5 V USB Type-C power and follow the screen. Each magnet step times out after 30 seconds.

Low-current power banks may switch off automatically. If this happens, use another regulated 5 V USB source.



1. Remove sphere; keep probe still.



2. Attach sphere squarely; do not move.



3. READY confirms start-up adaptation.

BOOT TIMEOUT / RAW MODE: restart and verify against a known reference.

MEASURING 10



Example live display

- Clean the probe nose, magnet sphere, and measuring surfaces.
- Seat the probe consistently and place the sphere on the opposite side of the material.
- Keep the probe square to the surface. Avoid side load, vibration, and sliding the magnet during a reading.
- Read after the value settles. W and S are drift references only; neither is permission or a requirement to measure.
- Check critical readings with a known reference.

ONE-BUTTON CONTROL 11

SHORT PRESS

Live: freeze or release the reading. When freezing, LUQER types the value through USB into the focused app, such as a text editor, LUQER Mapper, or spreadsheet. Menus: move to the next item. A short press also wakes the OLED.



Frozen value

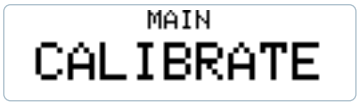
HOLD

Live: hold about 1.5 seconds to open the menu. Menus: hold about 1 second to select. For relative manual zeroing, see panel 22.



Zero confirmation

USER MENU 12



MAIN

- **INFO - firmware and serial number**
- **CALIBRATE**
  - **START - guided BASIC CAL**
  - **RESET - remove BASIC CAL**
  - **EXIT - return to MAIN**
- **UPDATE - reboot for firmware update**
- **ROTATION - RIGHT / LEFT / EXIT**
- **EXIT**

Short press cycles. Hold selects. EXIT returns to the previous level or live measurement.

BASIC CAL OVERVIEW 13

Use a clean, verified 3.00 mm shim. Keep the magnet attached throughout and allow the probe, shim, and workpiece to reach a stable temperature.



MAIN -> CALIBRATE -> START

- Capture zero before inserting the shim.
- Insert the verified 3.00 mm shim and capture it.
- Remove the shim, return to zero, and capture zero again.
- The instrument validates and saves automatically; follow any restart prompt.

DISPLAY SYMBOLS 14

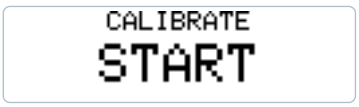


Magnet not detected

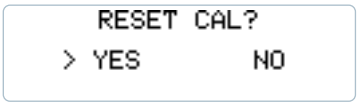
- S - combined thermal and ADC readiness criteria are met; reference only
- W - readiness criteria are not yet met; readings can still appear stable
- W / S do not enable, block, or define when a measurement may be taken
- C - BASIC CAL table is active
- M - manual zero is active
- AL - LUQER AL is enabled and collecting
- HOLD - the shown value is frozen
- -/- - beyond the supported display range
- NO MAGNET - the external magnet is not attracted by the probe
- 0.00 - values within about +/- 0.03 mm may be clamped to zero

CALIBRATE + RESET 15

CALIBRATE contains only the normal-user BASIC CAL controls shown below.



CALIBRATE -> START



CALIBRATE -> RESET

- START opens the three-capture BASIC CAL procedure.
- RESET disables BASIC CAL and restores the stored PRO calibration.
- After RESET, follow the normal REMOVE MAGNET / ATTACH MAGNET start-up prompts.
- EXIT returns to MAIN without changing calibration.

QUICK TROUBLESHOOTING 16

- Unstable reading: support the probe, sphere, cable, and workpiece and allow the value to settle. Do not use W / S as pass-fail instructions.
- NO MAGNET: the external magnet is not attracted by the probe. Check its placement and measuring distance.
- Unexpected offset: restart and repeat the start-up sequence, then verify with a known reference.
- No display: check the 5 V USB Type-C supply, cable, and connector.
- CAL FAILED / CHECK POSITION: clean the setup, confirm the 3.00 mm shim, and repeat all captures.
- Damaged housing, OLED, connector, button, nose, or magnet: stop use.

GOOD MEASUREMENT PRACTICE 17

- Use the same probe orientation, measuring location, and magnet placement for comparisons.
- Keep the probe nose perpendicular to the workpiece and avoid rocking or side pressure.
- Keep the 10 mm sphere clean. Ferrous dirt on the sphere changes contact and can scratch finished surfaces.
- Let the instrument and reference standards stabilize together before precision work.
- Take repeated readings and investigate unexpected spread rather than averaging an unstable setup.
- Record firmware version, serial number, operator, reference, and environment when traceability matters.

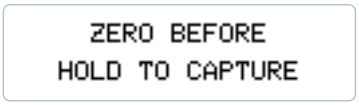
USB TYPE-C POWER 18

Use a stable, regulated 5 V USB supply and a sound USB Type-C cable. The cable and power source are not supplied.

- Insert the connector straight; do not twist or lever it sideways.
- Route the cable so it cannot pull the probe or disturb the reading.
- Disconnect by gripping the plug, not the cable.
- Do not connect a wet, dirty, damaged, or loose connector.
- Stop use if the connector becomes hot or power is intermittent.

When a live reading is frozen, LUQER types it as USB keyboard text into the focused app - for example a text editor, LUQER Mapper, or spreadsheet. Place the cursor in the intended field first; otherwise keep focus away from important documents.

BASIC CAL - ZERO BEFORE 19



CAPTURE FIRST ZERO

- Keep the 10 mm sphere attached at the normal zero position with no shim.
- Hold the probe square and keep the assembly still.
- Hold the button to capture when the setup is stable.

CAPTURED / NEXT: 3.00 MM confirms the first zero.

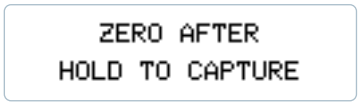
BASIC CAL - 3.00 mm 20



CAPTURE 3.00 mm

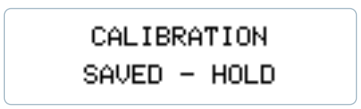
- Insert a clean, verified 3.00 mm shim without changing the probe orientation.
- Keep the sphere centred and fingers clear of the pinch point.
- Hold the button after the complete assembly is stable.
- CAPTURED / NEXT: ZERO confirms the shim point.

BASIC CAL - ZERO AFTER 21



CAPTURE FINAL ZERO

- Remove the shim and return the sphere to the same zero position.
- Keep the probe square and the sphere centred.
- Hold to capture after the assembly is stable.



Successful validation saves automatically.

If SAVED - HOLD appears, hold the button to restart.

MANUAL ZERO + NEGATIVE 22

Manual zero is a temporary relative offset. It does not rewrite the stored calibration.

- On the live screen, short press to freeze a valid reference reading.
- With HOLD shown, press and hold about 1.5 seconds until ZEROED appears.
- Live measurement resumes relative to the frozen reference.



Manual-zero confirmation

NEGATIVE VALUES



Example: 0.15 mm below manual zero

Below the reference, values are negative; above it, positive. Repeat to set a new reference. Manual zero is cleared by restart or calibration reload.

STORE IN THE CASE 23

- Disconnect USB power before packing.
- Place the probe in its fitted recess with the measuring nose oriented away from the magnet-storage-slot side.
- Secure the 10 mm sphere in the case's dedicated magnet storage slot so it cannot roll or snap onto the probe.
- Close the case only when every item is fully seated. Never force the lid.
- Do not store the case where the temperature can exceed 50 degrees C. Keep it away from impact, direct sunlight, and sensitive magnetic items.

CLEANING + CARE 24

- Wipe the enclosure and OLED with a soft cloth.
- Remove metal particles from the sphere and probe nose carefully before and after use.
- Do not immerse, spray liquid into openings, or use aggressive solvents.
- Do not drop, clamp, drill, grind, modify, or expose the probe to excessive heat.
- Inspect the nose, housing, button, OLED, USB-C socket, cable, and sphere regularly.
- After impact or repair, verify performance before returning the instrument to use.